SmartStock: A Web-Based Inventory, Sales, and Finance Management Information System for Salak Sari-Sari Store

Web Application Proposal

Area of Application: Transaction Processing System (TPS) and

Retail Business Operations Management System

Submitted to:

Everild Gerd A. Pablo, MBA

Instructor

Submitted by:

John Michael V. Mujar

Mary Ann S. Lumbria

Akisha Mekel C. San Miguel

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BACKGROUND OF THE STUDY

In today's fast-paced business environment, technology has become an essential tool for improving the efficiency and productivity of organizations. Many businesses, both big and small, are adopting digital systems to manage their daily operations. These computerized systems help reduce manual work, minimize errors, and provide quicker access to important information, enabling business owners to make smarter decisions.

Small retail businesses, like Salak Sari-Sari Store, often operate with limited staff and resources. Despite the growth of technology, many of these stores still rely on traditional manual methods, such as notebooks, paper records, or mobile phone notes, to track inventory, sales, and financial transactions. While these methods may be sufficient for small-scale operations, they become increasingly inefficient as the number of products and daily transactions grows.

Manual record-keeping demands significant time and effort. Tasks such as updating stock levels, calculating profits, monitoring expenses, and generating reports can become tedious and prone to errors. For instance, sales records may be incomplete, product quantities may be miscounted, and financial calculations may occasionally be inaccurate. Without an organized system, important information can easily be misplaced or overlooked, making it difficult for store owners to evaluate the overall performance of their business.

Salak Sari-Sari Store, the focus of this study, has been operating for approximately four to six years and currently offers around 101 to 200 different products. The store manages its operations entirely manually, requiring the store owner to constantly update inventory and sales records throughout the day. This process not only consumes three to four hours of the owner's time per inventory session but also increases the likelihood of errors, such as duplicate entries or missing information.

Another challenge faced by the store is the absence of an automated system that can track stock levels, categorize products, monitor prices, and alert the owner about items running low or nearing expiration. Monitoring financial performance is equally challenging since profit and expense

calculations are done manually. The lack of real-time information prevents the owner from making timely and well-informed business decisions.

Given these challenges, the adoption of a digital, web-based system can significantly improve the day-to-day operations of Salak Sari-Sari Store. By centralizing all records in a secure, organized database, the store owner can access up-to-date information about inventory, sales, and finances at any time. Additionally, a digital system can provide automated reports, dashboard summaries, and analytics, making it easier to track business performance, spot trends, and respond promptly to operational needs.

This study proposes the development of SmartStock: A Web-Based Inventory, Sales, and Finance Management Information System for Salak Sari-Sari Store. The system aims to provide a centralized platform for managing inventory, recording sales, tracking expenses, and generating business reports. With this system, the store owner can save time, reduce errors, and focus more on growing the business while making informed decisions that ensure sustainability and profitability.

PROBLEM STATEMENT

Effective management of inventory, sales, and financial records is crucial for the success of any retail business. In small stores like Salak Sari-Sari Store, these tasks are often handled manually due to limited resources and the absence of digital tools. While manual methods may work for a small scale, they become increasingly difficult as the number of products and daily transactions grows.

Currently, most operations are recorded using paper-based records and mobile phone notes. Updating product information, sales, and expenses manually takes three to four hours per inventory session, consuming time that could be used for other operational activities.

These manual processes introduce several challenges, including:

- Inventory tracking can be inaccurate, leading to difficulties in monitoring stock levels and identifying products that need restocking.
- Sales transactions are prone to data entry errors, incomplete records, and occasional duplicates.
- Calculating profit, tracking expenses, and generating financial summaries are time-consuming and prone to mistakes.
- Difficulty to identify Best-Selling and Poor-Selling Products.
- Difficulty to identify Expired and Non-Expired Products.
- There is no real-time visibility of sales performance, stock levels, or financial metrics, limiting the ability to make quick and informed decisions.
- Generating reports for management review or planning requires manual computation, which is inefficient and increases the risk of errors.

These challenges highlight the need for a web-based inventory, sales, and finance management system. Implementing SmartStock can help automate tasks, centralize data, reduce errors, and provide real-time insights, enabling the store owner to make informed, data-driven decisions.

OBJECTIVES OF THE STUDY

General Objective

The main objective of this study is to develop a Web-Based Inventory, Sales, and Finance Management Information System for Salak Sari-Sari Store that will help automate daily operations and improve the monitoring of inventory, sales, and financial data.

Specific Objectives

- To create a system that allows efficient and accurate management of product inventory for Salak Sari-Sari Store.
- To provide a digital platform for recording and monitoring daily sales transactions, reducing reliance on manual entry.

- To enable the store owner to record, categorize, and track expenses for better financial management.
- To generate automated reports and dashboard summaries for sales, expenses, and profit, providing real-time insights into business performance.
- To provide charts and analytics that highlight product performance, revenue trends, and low-stock alerts.
- To reduce manual errors and improve the accuracy of inventory, sales, and financial records.

SCOPE AND LIMITATIONS

Scope of the Study

This study focuses on the development of SmartStock, a web-based system designed to help manage the operations of Salak Sari-Sari Store.

The system will:

- Provide a secure login system for the store owner or administrator.
- Allow users to add, edit, delete, and search product records.
- Manage product information, including product name, category, price, and stock quantity.
- Record sales transactions and automatically calculate the total sales amount.
- Record store expenses and categorize them for financial monitoring.
- Generate dashboard summaries showing sales, expenses, and profit.
- Display charts showing monthly sales performance and product trends.
- Provide alerts for products that need restocking and nearly expired items.
- Allow customization options such as system color themes.
- Be accessible through a web browser using a computer, laptop, tablet, or smartphone where the system is deployed.

- The system interface will be mobile responsive, allowing the layout to automatically adjust to different screen sizes for better usability on mobile devices.

Limitations of the Study

Despite its benefits, the system has the following limitations:

- The system is designed for Salak Sari-Sari Store only.
- The system will be web-based only and will not include a mobile application.
- All product, sales, and expense information must still be manually entered by the user.
- The system will not include supplier management or barcode scanning features.
- Reports cannot be printed directly from the system.
- The system can operate in two ways: online (requiring internet access) or offline (local deployment), which may limit accessibility depending on the chosen setup.

TARGET CLIENT AND USER

Target Client

The target client of the proposed system is Salak Sari-Sari Store, a local retail store that currently uses manual methods for recording inventory, sales, and expenses.

System User

- Store Owner / Administrator
  - Managing product inventory
  - Recording sales transactions
  - Monitoring expenses and profits
  - Viewing dashboard reports and charts
  - Managing account settings and system preferences

PROPOSED SYSTEM FEATURES

1. Operational Features (Web System)

1.1 User Login System

The system shall provide a secure login mechanism to ensure that only authorized users can access the system.

1.1.1 The system shall allow the user to log in using a username and password.

1.1.2 The system shall verify the entered credentials before granting access to the system.

1.1.3 The system shall allow the user to log out of the system after finishing their tasks.

1.2 Inventory Management

The system shall provide a module for managing product inventory to help monitor product information and stock levels.

1.2.1 The system shall allow the user to add new products including product name, category, price, and stock quantity.

1.2.2 The system shall allow the user to edit existing product information when changes are needed.

1.2.3 The system shall allow the user to delete product records that are no longer available in the store.

1.2.4 The system shall automatically update stock levels when products are sold.

1.3 Product Search and Filtering

The system shall provide search and filtering functions to help users quickly find products.

1.3.1 The system shall allow the user to search products by name or category.

1.3.2 The system shall allow the user to filter products based on category or stock status.

1.3.3 The system shall display matching results instantly to make product

searching easier.

1.4 Sales Recording

The system shall provide a module for recording daily sales transactions.

1.4.1 The system shall allow the user to select a product and input the quantity sold.

1.4.2 The system shall record sales details such as product name, quantity, and date of transaction.

1.4.3 The system shall automatically reduce the product stock quantity after a sale is recorded.

1.5 Automatic Sales Calculation

The system shall automatically compute sales totals to reduce manual calculation errors.

1.5.1 The system shall automatically calculate the total price of products sold based on quantity and unit price.

1.5.2 The system shall display the total amount of each transaction.

1.5.3 The system shall store the sales data for reporting and monitoring.

1.6 Expense Management

The system shall allow the user to record and manage store expenses.

1.6.1 The system shall allow the user to add new expense records such as utilities, restocking, or other business expenses.

1.6.2 The system shall allow the user to edit or delete expense records when needed.

1.6.3 The system shall categorize expenses for easier financial tracking.

1.7 Stock Monitoring

The system shall monitor inventory levels and notify the user when products are running low and nearly expiring.

1.7.1 The system shall automatically detect products that have low stock

quantities.

1.7.2 The system shall display low stock alerts on the dashboard.

1.7.3 The system shall help the user identify products that need restocking.

1.7.4 The system shall Alert for products nearing expiration.

1.8 Account Settings

The system shall allow the user to manage their account information.

1.8.1 The system shall allow the user to change their username and password.

1.8.2 The system shall provide a password reset option in case the user forgets their password.

1.8.3 The system shall allow the user to update their personal account settings.

1.9 Appearance Customization

The system shall allow users to customize the interface based on their preferences.

1.9.1 The system shall allow users to change the system color theme.

1.9.2 The system shall save the user's preferred appearance settings.

1.10 Mobile Responsive Interface

The system shall provide a mobile responsive interface to ensure usability across different devices.

1.10.1 The system shall automatically adjust its layout depending on the screen size of the device.

1.10.2 The system shall support viewing and basic system functions using smartphones and tablets.

1.10.3 The system shall maintain readability and usability on smaller screens.

2. MIS Features (Management Information System)

2.1 Dashboard Overview

The system shall provide a dashboard that displays important information about store operations.

2.1.1 The dashboard shall display total sales revenue.

2.1.2 The dashboard shall show total expenses recorded in the system.

2.1.3 The dashboard shall display net profit based on recorded sales and expenses.

2.1.4 The dashboard shall show low stock alerts for products that need restocking.

2.2 Sales Reports

The system shall generate reports that summarize sales transactions.

2.2.1 The system shall generate daily sales reports.

2.2.2 The system shall generate monthly sales reports.

2.2.3 The reports shall display total sales amount and number of products sold.

2.3 Product Performance Monitoring

The system shall analyze sales data to identify product performance.

2.3.1 The system shall identify top-selling products based on sales data.

2.3.2 The system shall identify slow-moving products that have low sales activity.

2.3.3 The system shall display product performance information on the dashboard.

2.4 Financial Reports

The system shall generate financial summaries for monitoring business performance.

2.4.1 The system shall display total revenue generated from sales.

2.4.2 The system shall display total expenses recorded in the system.

2.4.3 The system shall calculate and display net profit.

2.5 Charts and Analytics

The system shall provide visual charts and graphs for easier data analysis.

2.5.1 The system shall display charts showing monthly sales trends.

2.5.2 The system shall display charts showing product performance.

2.5.3 The system shall provide graphical summaries of sales and financial data.

INITIAL SYSTEM OVERVIEW DIAGRAM

Context Diagram

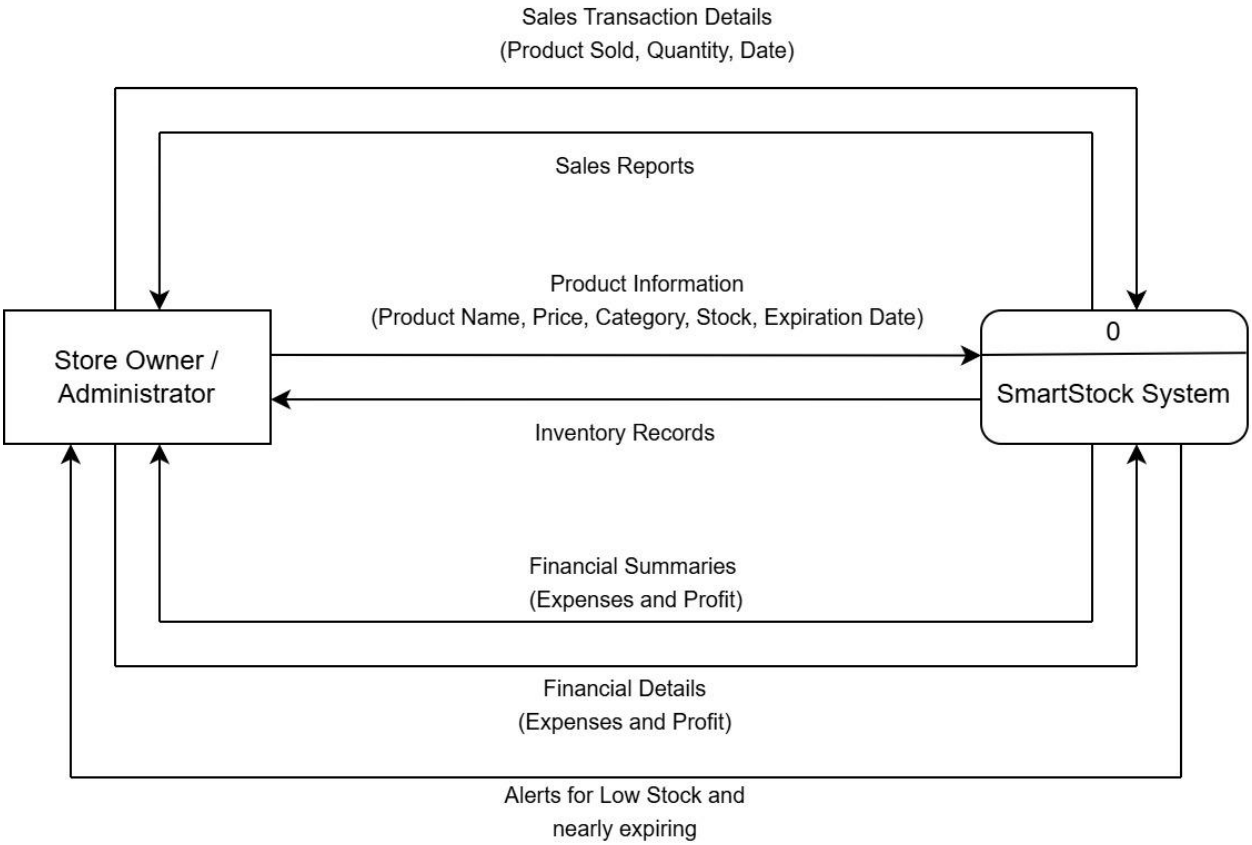

Figure 1: Context Diagram of SmartStock System (Draw.io)

The Context Diagram shows how the SmartStock system interacts with its main user, the store owner or administrator of Salak Sari-Sari Store. The owner enters product details, sales transactions, and financial data into the system. SmartStock then processes this information and produces outputs like inventory reports, sales summaries, financial summaries (including expenses and profit), dashboard charts, and alerts for products nearing expiration. These outputs help the owner monitor stock levels, track sales performance, identify items that are close to expiring, and make informed decisions to manage the business efficiently.

Level 0 Data Flow Diagram

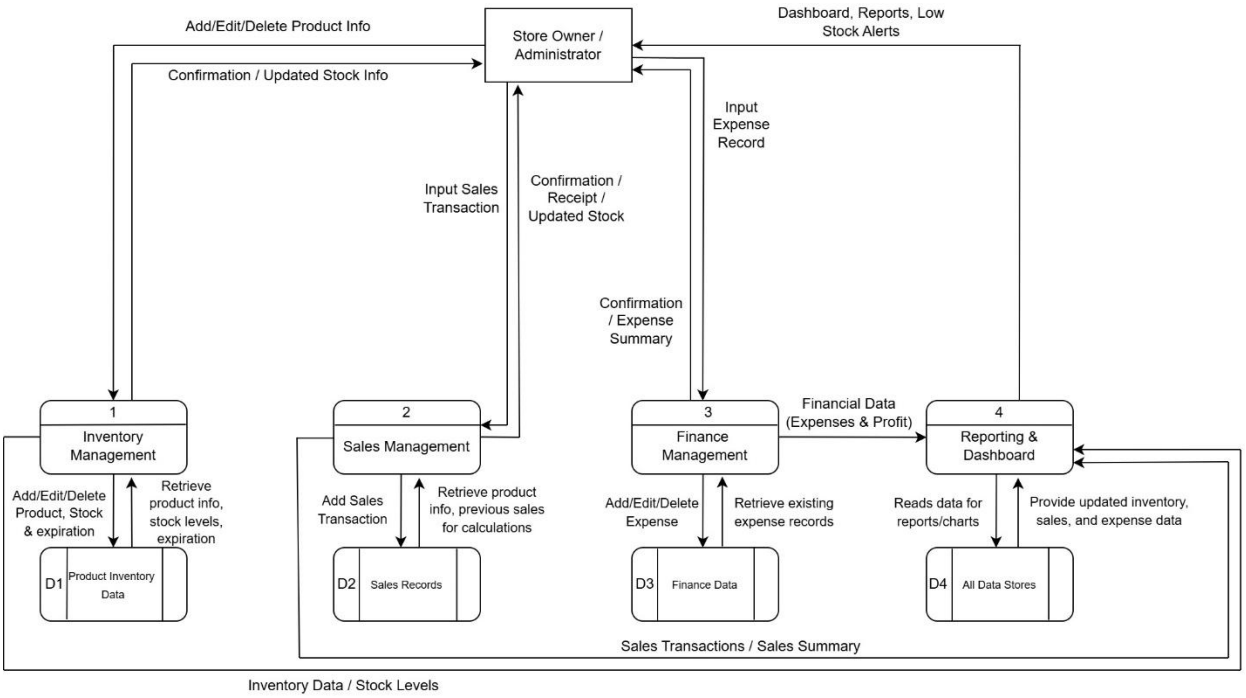

Figure 2: Level 0 Data Flow Diagram of the SmartStock System (Draw.io)

The Level 0 DFD of SmartStock shows how the store owner of Salak Sari-Sari Store interacts with the system and how data flows within it. The owner enters product information, sales transactions, and financial data into the system. These inputs are handled by three main modules: Inventory Management, Sales Management, and Finance Management, which store and update the data in their respective databases.

The stored data is then used by the Reporting & Dashboard module to generate sales reports, financial summaries (including expenses and profit), charts, and low-stock alerts, which are sent back to the store owner. Data flows in both directions where needed: the system retrieves information from the databases to provide outputs and confirmations, while the owner adds new inputs.

This diagram shows how SmartStock collects, processes, stores, and presents information to help the store owner make informed decisions and manage the store more efficiently.

Process Flow Diagram

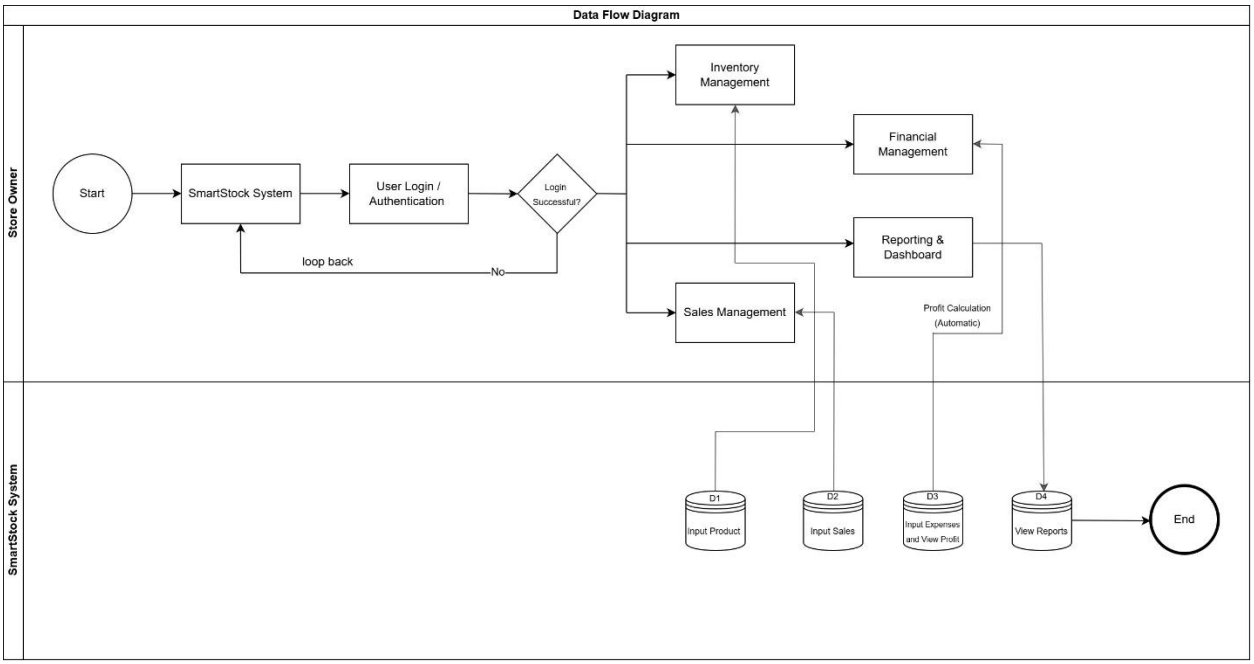

Figure 3: Process Flow Diagram of the SmartStock System (Draw.io)

The Process Flow Diagram shows how the SmartStock system operates and how the Store Owner interacts with it. The process begins when the Store Owner starts the system and logs in. Once the login is successful, the user can manage inventory, record sales, and input financial data (such as expenses) through the system modules. These activities update the system's records and databases. The system then processes the stored information, automatically calculates profit, and generates reports, dashboard summaries, and alerts, which are displayed to the Store Owner. The process ends after the user views the generated reports and information from the system.

CONCLUSION

The assessment of Salak Sari-Sari Store shows that relying on manual methods for inventory, sales, and financial records creates challenges such as time-consuming processes, errors, and delays in monitoring performance.

To address these issues, the proposed SmartStock: Web-Based Inventory, Sales, and Finance Management Information System provides a centralized platform to efficiently manage products, sales transactions, expenses, and generate summarized reports.

With features like dashboards and visual charts, the system allows the store owner to track sales trends, identify top-selling and slow-moving products, monitor stock levels, and check for expired items, as well as monitor financial performance. This access to real-time information enables informed decision-making and improved operational efficiency.

Overall, SmartStock aims to simplify daily operations, reduce manual work, and enhance accuracy, allowing Salak Sari-Sari Store to transition to a more organized and reliable digital system.